MIL-F-50864B(AR)
07 MARCH 1988
SUPERSEDING
MIL-F-50864A(MU)
29 March 1976

MILITARY SPECIFICATION
FUZE, PIBD, M549; LESS SPITBACK METAL PARTS AND LOADING, ASSEMBLING AND PACKING

This specification is approved for use within the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the metal parts and loading, assembling and packing for one type of fuze designated as Fuze, PIBD, M549; Less Spitback.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issue of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-P-116 - Preservation, Packaging, Method of
MIL-C-5541 - Chemical Conversion Coatings on Aluminum and Aluminum Alloys
MIL-A-8625 - Anodic Coatings for Aluminum and Aluminum Alloys
MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification for
MIL-D-50865 - Detonator, Stab, M59

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Armament Munitions and Chemical Command, Attn: AMSMC-QA, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 1390
AMSC N/A
DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:03Z
Check the source to verify that this is the current version before use.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-331 - Fuze and Fuze Components, Environmental and Performance Tests for
MIL-STD-453 - Inspection, Radiographic

2.1.2 Other Government documents and drawings. The following other Government documents and drawing form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of solicitation.

DRAWINGS (See 6.5)

US ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

9287860 - Fuze, PIBD, M549
9287861 - Fuze, PIBD, M549, Less Spitback
9287859 - Box, Fiberboard, Packing for Fuze, PIBD, M549, Less Spitback
9287862 - Spitback Assembly
9349828 - Test Vehicle, w/Crtg Case M169 for M549 Fuze
9349971 - Cartridge, 40mm, Test Vehicle, for M549 Fuze
12624504 - Cup, Spitback

(Copies of other Government documents and drawings required by contractors in connection with specified acquisition functions should be ovtained from the contracting activity or as directed by the contracting officer.)

CODE OF FEDERAL REGULATIONS (CFR)

TITLE 49 - Transportation, Parts 100-199

(The Code of Federal Regulations is available from the Superintent of Documents, US Goverment Printing Office, Washington, D.C. 20402. Orders should specify, "49 CFR 100-199 (lastest revision)).

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of solicitation.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:03Z
Check the source to verify that this is the current version before use.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM B 117-73 - Standard Method of Salt Spray (Fog) Testing

(Application for copies of ASTM publications, should be addressed to: The American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing, however, shall supersede applicable laws and regulations unless a specified exemption has been obtained.

3. REQUIREMENTS

3.1 Material. Materials shall be in accordance with applicable drawings and specifications.

3.2 Parts. The parts shall comply with all requirements specified on Drawing (Dwg) 9287861, all associated drawings, and with all requirements specified in applicable specifications.

3.3 X-Ray examination of fuze assembly. The fuze assembly shall be x-rayed and shall not be armed or partially armed, have any safety feature missing (setback pin, setback spring or safety spring), safety feature not in its proper position (i.e. setback pin not seated or safety spring not engaged in teeth of rotor gear) or pinion assembly, verge assembly or o-ring missing or drive screw missing or not seated.

3.4 Transportation vibration. After successful completion of x-ray examination, the fuze assembly shall be subjected to the transportation vibration test. It shall not function during or after the test and after the test it shall not be armed, partially armed, have any safety feature not in its proper position or have the drive screw not seated. After successful completion of the test, the fuze assembly shall be subjected to and successfully pass either the air pressure, functioning, or non-functioning test.

3.5 Jolt. After successful completion of x-ray examination, the fuze assembly shall be subjected to the jolt test. It shall not function during or after the test and after the test it shall not be armed, partially armed, have any safety feature not in its proper position or have the drive screw not seated.

3.6 Jumble. After successful completion of x-ray examination, the fuze assembly shall be subjected to the jumble test. It shall not function during or after the test and after the test it shall not be armed, partially armed, have any safety feature not in its proper position or have the drive screw not seated.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:03Z
Check the source to verify that this is the current version before use.

3.7 Functioning. The fuze assembly shall function high order as evidenced by the emittance of a visual (smoke or flash) or audible signature when it strikes the specified impact medium.

3.8 Non-functioning. The fuze assembly shall not function as evidenced by the lack of a visual (smoke and flash) and audible signature when it strikes the specified target.

3.9 Premature burst. The fuze assembly shall not function in the gun tube or in uninterrupted flight during the functioning or non-functioning test.

3.10 First article inspection. This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.11 Quality program requirements. The contractor's quality program plan shall include detailed documentation for all critical defect material handling procedures and process controls to be used in the manufacture of the fuze. The documentation shall specically address the requirements for fuzes in MIL-A-48078 and will be in accordance with the approved Contract Data Requirements List and data item description DI-R-5297 when so specified in the contract (see 6.3). This documentation shall be submitted through the contracting officer to the technical agency for review and the contractor shall not begin production without approval thereof from the contracting officer. Changes subsequent to approval shall also be documented and submitted to the Government for review and approval prior to implementation.

3.12 Workmanship. All parts and assemblies shall be fabricated, loaded and assembled in a thorough and workmanlike manner. They shall be free of burrs, sharp edges, cracks, dirt, grease, rust and other foreign matter. The cleaning method used shall not be injurious to any parts, nor shall the parts be contaminated by the cleaning agents. Exterior surface coatings shall be continuous; however, a few light scratches not exposing base material may be permitted. Rack marks on the ogive are permissible. All required marking and stamping shall be neat and sharply defined.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:03Z
Check the source to verify that this is the current version before use.

conform to prescribed requirements. In addition, unless specified herein or in the contract the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the government to acceptance of defective material.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection

b. Quality conformance inspection

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.3.2. The first article sample shall consist of the following items in sample quantities as indicated.

PART DESCRIPTION	DRAWINGS	QUANTITY
Escapement Housing	9287867	25
Verge Arbor	9287880	25
Pinion	9287873	25
Ogive	9287888	25
Firing Pin	9287886	25
Setback Pin	9287882	25
Safety Spring	9287870	125
Anti-Creep Spring	9287881	125
Setback Spring	9287883	125
Starwheel	8886370	25
Pinion Assembly	9287872	125
Verge Assembly	9287877	125
Top Plate Assembly	9287884	125
Safety Spring Assembly	9287869	125
Rotor and Gear Assembly	9205410	125

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:03Z
Check the source to verify that this is the current version before use.

PART DESCRIPTION	DRAWINGS	QUANTITY
Ogive Assembly	9287887	125
Ogive Assembly with O-Ring Installed	9287887 and MS 29513-214	125
Escapement Assembly Less Setback Pin and Setback Spring	9287866 (less 9287882 and 9287883)	125
Escapement Assemby	9287866 (see 4.5.15 for configuration)	125
Escapement Assembly	9287866	125
Escapement Assembly and Ogive assembly with O-ring installed	9287866, 9287887 and MS 29513-214	125
Fuze, PIBD, M549, Less Spitback	9287861	508

4.3.2 Inspections to be performed. First article components and assemblies shall be subjected to any or all of the examinations and tests specified in Table 1 and to any or all requirements of the applicable drawings.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:03Z
Check the source to verify that this is the current version before use.

TABLE I - First article inspection

CLASSIFICATION OF DEFECTS & TESTS MIL-F-50864B (AR)

PARAGRAPH	TITLE: Fuze, PIBD, M549 Less Spitback and Components		SHEET 1 OF 4		DRAWING NUMBER: See Below NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Housing, Escapement (Dwg. 9287867)				
	Examination for Defects	25		3.2	4.4.2.1
	Arbor, Verge (Dwg. 9287880)				
	Examination for Defects	25		3.2	4.4.2.2
	Pinion (Dwg. 9287873)				
	Examination for Defects	25		3.2	4.4.2.3
	Ogive (Dwg. 9287888)				
	Examination for Defects	25		3.2	4.4.2.4
	Salt Spray	4 1/		3.2	4.5.1
	Firing Pin (Dwg. 9287886)				
	Examination for Defects	25		3.2	4.4.2.5
	Setback Pin (Dwg. 9287882)				
	Examination for Defects	125		3.2	4.4.2.6
	Safety Spring (Dwg. 9287870)				
	Examination for Defects	125		3.2	4.4.2.7
	Load requirements	125		3.2	4.5.2
	Anti-Creep Spring (Dwg. 9287881)				
	Examination for Defects	125		3.2	4.4.2.8
	Load requirements	125		3.2	4.5.2

NOTES: 1/ Samples taken from quantity of 25.

Source: https://assist.dla.mil -- Downloaded: 20260-26T04:03Z
Check the source to verify that this is the current version before use.

TABLE I - First article inspection

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-50864B (AR)

PARAGRAPH	TITLE Fuze, PIBD, M549 Less Spitback and Components		SHEET 2 OF 4		DRAWING NUMBER See Below NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Setback Spring (Dwg. 9287883)				
	Examination for Defects	125		3.2	4.4.2.9
	Push test of simulated setback pin	125		3.2	4.5.3
	Starwheel (Dwg. 8886370)				
	Examination for Defects	25		3.2	4.4.2.10
	Pinion Assembly (Dwg. 9287872)				
	Examination for Defects	125 1/		3.2	4.4.2.11
	Push off of starwheel	125 1/		3.2	4.5.4
	Torque of starwheel (when applicable)	125 1/		3.2	4.5.5
	Verge Assembly (Dwg. 9287877)				
	Examination for Defects	125		3.2	4.4.2.12
	Push off of verge arbor	125		3.2	4.5.6
	Top Plate Assembly (Dwg. 9287884)				
	Examination for Defects	125		3.2	4.4.2.13
	Push out of firing pin	125		3.2	4.5.7
	Safety Spring Assembly (Dwg. 9287869)				
	Examination for Defects	125		3.2	4.4.2.14

NOTES: 1/ Same units.

AMSMC Form 1570, 1 Feb 85

Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:03Z
Check the source to verify that this is the current version before use.

TABLE I - First article inspection

CLASSIFICATION OF DEFECTS & TESTS MIL-F-50864B (AR)

PARAGRAPH	TITLE Fuze, PIBD, M549 Less Spitback and Components		SHEET 3 OF 4		DRAWING NUMBER See Below NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Rotor and Gear Assembly (Dwg. 9205410)				
	Examination for Defects	125		3.2	4.4.2.15
	Stake of detonator	125		3.2	4.5.8
	Push out test of rotor	125		3.2	4.5.9
	Ogive Assembly (Dwg. 9287887)				
	Examination for Defects	125		3.2	4.4.2.16
	Push test of actuator	125		3.2	4.5.10
	Ogive Assembly with O-Ring Installed (Dwg. 9287887 and MS 29513-214)				
	Examination for Defects	125		3.2	4.4.2.17
	Escapement Assembly Less Setback Pin and Setback Spring (Dwg. 9287866 less 9287882 and 9287883)				
	Examination for Defects	125		3.2	4.4.2.18
	Arming time, minimum	125		3.2	4.5.12
	Arming time, maximum	125		3.2	4.5.12
	Non-arming test	125		3.2	4.5.13
	Escapement Assembly (Dwg. 9287866- See 4.5.15 for configuration)				
	Stake of top plate assembly	125		3.2	4.5.15

NOTES:

AMSMC Form 1570, 1 Feb 85 Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:03Z
Check the source to verify that this is the current version before use.

TABLE I - First article inspection

CLASSIFICATION OF DEFECTS & TESTS MIL-F-50864B (AR)

PARAGRAPH	TITLE: Fuze, PIBD, M549 Less Spitback and Components		SHEET 4 OF 4		DRAWING NUMBER: See Below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Escapement Assembly (Dwg. 9287866)				
	Examination for Defects	125		3.2	4.4.2.19
	Escapement Assembly and O-Ring Assembled into Ogive Assembly (Dwg. 9287887, MS 29513-214 and 9287866)	125		3.2	4.4.2.20
	Fuze, PIBD, M549 Less Spitback (Dwg. 9287861)				
	Examination for Defects	508		3.2	4.4.2.21
	X-ray examination of fuze Assembly	508		3.3	4.5.16
	Transportation Vibration	400		3.4	4.5.17
	Jolt	12		3.5	4.5.18
	Jumble	96		3.6	4.5.19
	Air pressure	125		3.2	4.5.20
	Functioning			3.7	4.5.21
	Coarse sand	150	Acc 5 Rej 6		4.5.21.1
	1/2 or 5/8 Inch Celotex	150	Acc 5 Rej 6		4.5.21.2
	Non-Functioning	100	Acc 0 Rej 1	3.8	4.5.22
	Premature burst	400	Acc 0 Rej 1	3.9	4.5.23

NOTES:

AMSMC Form 1570, 1 Feb 85 Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:03Z
Check the source to verify that this is the current version before use.

4.3.3 Rejection. The first article quantity shall be rejected upon failure of any component or assembly in the sample to comply with any visual or dimensional characteristics or upon failure to comply with the criteria for any test. The Government reserves the right to terminate its inspection whenever a component or assembly fails to comply with any visual or dimensional characteristics or components or assemblies fail to comply with the criteria for any test.

4.4 QUALITY CONFORMANCE INSPECTION.

4.4.1 Inspection lot formation. The term "inspection lot" is defined as a homogeneous collection of units of product from which a representative sample is drawn or which is inspected 100 percent to determine conformance with applicable requirements. Units of product selected for inspection shall represent only the inspection lot from which drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. The inspection lot shall comply with the provisions for submission of product as specified in MIL-STD-105. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein, regardless of the type of inspection procedure which is being applied to determine conformance with requirements. Lot numbering, as required, shall be in accordance with MIL-STD-1168. A lot shall consist of:

a. Detonators from one interfix lot series and from one manufacturer.

b. Metal parts from one interfix lot series and from one manufacturer.

4.4.2 Examination. Unless otherwise specified in the classification of defects and test tables, sampling plans for the individual major and minor defects listed shall be in accordance with MIL-STD-105, Inspection Level II. One hundred percent examination shall be performed for all critical defects.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:03Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.1	Housing, Escapement	1 OF 1	9287867
			NEXT HIGHER ASSEMBLY 9287866

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Diameter of drive screw hole, maximum		0.40%		Gage
102	Diameter of thru hole for smallest diameter of setback pin, maximum		0.40%		Gage
Minor					
201	Lubricant missing		0.65%		Visual
202	Poor workmanship		0.65%		Visual

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:03Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9287880
4.4.2.2	Arbor, Verge				NEXT HIGHER ASSEMBLY 9287877
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major 101	Bearing diameter (2)		0.40%	3.2	Gage
Minor 201	Poor workmanship		0.65%	3.12	Visual

NOTES:

AMSMC Form 1570, 1 Feb 85

Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:03Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-50864B (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9287873
4.4.2.3	Pinion				NEXT HIGHER ASSEMBLY 9287872
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major 101	Bearing diameter (2)		0.40%	3.2	Gage
Minor 201	Poor workmanship		0.65%	3.12	Visual

NOTES:

AMSMC Form 1570, 1 Feb 85

Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:03Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9287888
4.4.2.4	Ogive				NEXT HIGHER ASSEMBLY 9287887
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Wall thickness (average of 3 readings)		0.40%	3.2	Gage
102	Spherical radius		0.40%	3.2	Gage
103	Salt Spray	4	1/	3.2	4.5.1
Minor					
201	Poor workmanship		0.65%	3.12	Visual

NOTES: 1/ Rejection shall be in accordance with MIL-C-5541 or MIL-A-8625, as applicable.

Source: https://assist.dla.mil - Downloaded: 2026-10-26T04:03Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9287886
4.4.2.5	Firing Pin				NEXT HIGHER ASSEMBLY 9287884
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Profile of firing pin point		0.40%	3.2	Gage
102	Length from point to rear shoulder		0.40%	3.2	Gage
Minor					
201	Poor workmanship		0.65%	3.12	Visual

NOTES:

AMSMC Form 1570, 1 Feb 85

Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

Source: https://assist.dla.mil -- Downloaded: 2026-02-26T04:03Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

MIL-F-50864B (AR)

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9287882
4.4.2.6	Setback Pin				NEXT HIGHER ASSEMBLY 9287866
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical 1	Length of smallest diameter, min.		100%	3.2	Gage
Major 101	Smallest diameter		0.40%	3.2	Gage
Minor 201	Poor workmanship		0.65%	3.12	Visual

NOTES:

AMSMC Form 1570, 1 Feb 85 Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:03Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-50864B (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.7	Safety Spring	1 OF 1	9287870
			NEXT HIGHER ASSEMBLY 9287869

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major 101	Load requirement at each position	125	Acc-1 REJ-2	3.2	4.5.2
Minor 201	Poor workmanship		0.65%	3.12	Visual

NOTES:

AMSMC Form 1570, 1 Feb 85

Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:03Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-50864E (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9287881
4.4.2.8	Anti-creep Spring				NEXT HIGHER ASSEMBLY 9287866
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major 101	Load requirement at each position	125	Acc-1 Rej-2	3.2	4.5.2
Minor 201	Poor workmanship		0.65%	3.12	Visual

NOTES:

AMSMC Form 1570, 1 Feb 85

Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

Source: https://assist.dla.mil - Downloaded: 2026-10-26T04:03Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-50864B (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9287883
4.4.2.9	Setback Spring				NEXT HIGHER ASSEMBLY 9287866
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Push test of simulated setback pin	125	Acc-1 Rej-2	3.2	4.5.3
102	Diameter between tangs		0.40%	3.2	Gage
103	Tang broken, bent or missing		0.40%	3.2	Visual
104	Total length, min		0.40%	3.2	Gage
105	Diameter across legs		0.40%	3.2	Gage
106	Leg broken, bent or missing		0.40%	3.2	Gage
Minor					
201	Total length, max.		0.65%	3.2	Gage
202	Poor workmanship		0.65%	3.12	Visual

NOTES:

AMSMC Form 1570, 1 Feb 85

Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:03Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-50864B (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 8886370
4.4.2.10	Starwheel				NEXT HIGHER ASSEMBLY 9287872
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major 101	Starwheel thickness		0.40%	3.2	Gage
Minor 201	Poor Workmanship		0.65%	3.12	Visual

NOTES:

AMSMC Form 1570, 1 Feb 85 Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

Source: https://assist.dla.mil - Downloaded: 2026-10-26T04:03Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-50864B (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9287872
4.4.2.11	Pinion Assembly				NEXT HIGHER ASSEMBLY 9287866
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Push-off of starwheel	125	Acc-1 Rej-2	3.2	4.5.4
102	Torque of starwheel (when applicable)	125	Acc-1 Rej-2	3.2	4.5.5
103	Perpendicularity of starwheel with bearing diameter		0.40%	3.2	Gage
104	Concentricity of starwheel with bearing diameter		0.40%	3.2	Gage
Minor					
201	Poor workmanship		0.65%	3.12	Visual

NOTES:

AMSMC Form 1570, 1 Feb 85 Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:03Z
Check the source to verify that this is the current version before use.

POSTAGE AND FEES PAID DEPARTMENT OF THE ARMY

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:03Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-50864B (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9287877
4.4.2.12	Verge Assembly				NEXT HIGHER ASSEMBLY 9287866
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Push off of verge arbor	125	Acc-1 Rej-2	3.2	4.5.6
102	Thickness of verge and verge weight, max.		0.40%	3.2	Gage
103	Perpendicularity of verge with bearing diameter		0.40%	3.2	Gage
104	Length from verge to end of verge arbor		0.40%	3.2	Gage
Minor					
201	Lubricant missing		0.65%	3.2	Visual
202	Poor workmanship		0.65%	3.12	Visual

NOTES:

AMSMC Form 1570, 1 Feb 85 Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-50864B (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.13	Top Plate Assembly		9287884
			NEXT HIGHER ASSEMBLY 9287866

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Push out of firing pin	125	Acc-1 Rej-2	3.2	4.5.7
102	Protrusion of firing pin point, max.		0.40%	3.2	Gage
103	Firing pin point damaged		0.40%	3.2	Visual
Minor					
201	Lubricant missing		0.65%	3.2	Visual
202	Poor workmanship		0.65%	3.12	Visual

NOTES:

AMSMC Form 1570, 1 Feb 85

Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:03Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9287869
4.4.2.14	Safety Spring Assembly				NEXT HIGHER ASSEMBLY 9287866
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Weight loose		0.40%	3.2	Manual
102	Distance from weight to end of spring		0.40%	3.2	Gage
Minor					
201	Poor workmanship		0.65%	3.12	Visual

NOTES:

Source: https://assist.dla.mil - Downloaded: 2026-10-26T04:03Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-50864B (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9205410
4.4.2.15	Rotor and Gear Assembly				NEXT HIGHER ASSEMBLY 9287866
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Stake of detonator	125	Acc-1 Rej-2	3.2	4.5.8
102	Detonator inverted		0.40%	3.2	Visual
103	Detonator above flush		0.40%	3.2	Gage
104	Crack on detonator face		0.40%	3.2	Visual
105	Push out test of rotor	125	Acc-1 Rej-2	3.2	4.5.9
Minor					
201	Lubricant missing		0.65%	3.2	Visual
202	Poor workmanship		0.65%	3.12	Visual

NOTES:

AMSMC Form 1570, 1 Feb 85

Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:03Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER 9287887
4.4.2.16	Ogive Assembly		NEXT HIGHER ASSEMBLY 9287861

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Push test of actuator	125	Acc-1 Rej-2	3.2	4.5.10
102	Length from internal shoulder to actuator		0.40%	3.2	Gage
103	Pitch diameter of thread		0.40%	3.2	Gage
104	True position of large outside diameter with pitch diameter of thread		0.40%	3.2	Gage
Minor					
201	Thread length		0.65%	3.2	Gage
202	Diameter of thread undercut		0.65%	3.2	Gage
203	Width of thread undercut		0.65%	3.2	Gage
204	Color incorrect		0.65%	3.2	Visual
205	Protective finish with bare spot except rack marks		0.65%	3.12	Visual
206	Poor workmanship		0.65%	3.12	Visual

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:03Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET		DRAWING NUMBER
4.4.2.17	Ogive Assembly with O-Ring Installed		1 OF 1		9287887
					NEXT HIGHER ASSEMBLY 9287861
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical 1	O-ring missing 1/		100%	3.2	Automatic Inspection Equipment (AIE)/4.5.11
Major	None defined				
Minor	None defined				

NOTES: 1/ After successful completion of this inspection no subsequent operations which invert the ogive assembly shall be performed.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:03Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-50864B (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.18	Escapement Assembly Less Setback Pin and Setback Spring		9287866 1/
			NEXT HIGHER ASSEMBLY 9287861

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Arming time, minimum 2/		100%	3.2	4.5.12
2	Non-arming test (safety test) 3/		100%	3.2	4.5.13
3	Setback pin hole blocked 4/		100%	3.2	AIE/4.5.14
4	Safety spring end not in contact with root diameter surface or root of sector gear 5/		100%	3.2	AIE/4.5.14
5	Drivescrew missing or not seated		100%	3.2	AIE/4.5.14
Major					
101	Arming time, max. 2/		100%	3.2	4.5.12
102	Anti-creep spring missing		100%	3.2	Visual or AIE/4.5.14
103	Stake of top plate assembly	125	Acc-1 Rej-2	3.2	4.5.15

NOTES: 1/ Excluding Drawings 9287882 and 9287883
2/ These tests shall be performed concurrently
3/ This test shall be performed after successful completion of the arming time test
4/ This inspection shall be performed after successful completion of the non-arming test
5/ This inspection shall be performed after successful completion of the inspection for Critical 3, "Setback pin hole blocked"

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:03Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.19	Escapement Assembly				9287866
					NEXT HIGHER ASSEMBLY 9287861
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Setback pin not adjacent to rotor		100%	3.2	AIE/4.5.14 1/
Major	None defined				
Minor					
201	Poor workmanship		0.65%	3.12	Visual

NOTES: 1/The AIE shall sense the presence and minimum length of protusion into the escapement mechanism cavity of the smallest diameter of the setback pin.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:03Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-50864B (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.20	Escapement Assembly and O-Ring Assembled into Ogive Assembly		9287866 1/
			NEXT HIGHER ASSEMBLY 9287861

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Setback pin missing 2/		100%	3.2	AIE/4.5.1]
2	Setback spring missing 2/		100%	3.2	AIE/4.5.1]
Major	None defined				
Minor	None defined				

NOTES: 1/ Including drawings 9287887 and MS 29513-214
2/ These inspections may be performed at the escapement assembly level (4.4.2.19) providing acceptable escapement assemblies are automatically transferred to acceptable ogive assemblies with o-rings installed (4.4.2.17).

AMSMC Form 1570a, 1 Apr 85

Replaces DRDAR-QA Form 160a, 1 Jun 83, which may not be used

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:03Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 1 OF 3	DRAWING NUMBER
4.4.2.21	Fuze, PIBD, M549, Less Spitback		9287861
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Rotor in armed or partially armed		100%	3.2	Visual or AIE/4.5.14
2	X-ray examination of fuze assembly	318 1/	Acc-0 Rej-1	3.3	4.5.16
3	Transportation vibration	210	Acc-0 Rej-1	3.4	4.5.17
4	Jolt	12	Acc-0 Rej-1	3.5	4.5.18
5	Jumble	96	Acc-0 3/ Rej-1	3.6 2/	4.5.19

NOTES: 1/ Assemblies that are to be subjected to transportation vibration, jolt and jumble tests.
2/ If a detonator initiates during this test, the failure shall be considered a "Major" defect if the fuze is in the unarmed position.
3/ This accept/reject criteria also applies to any fuze that has a detonator initiate with the fuze in the unarmed position.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:03Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-50864B (AR)

PARAGRAPH	TITLE			SHEET 2 OF 3		DRAWING NUMBER 9287861
4.4.2.21	Fuze, PIBD, M549, Less Spitback					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST		NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Major						
101	Air pressure	80	1/	Acc-3 Rej-4	3.2	4.5.20
102	Functioning - coarse sand	100	1/2/	Acc-3 Rej-4	3.7	4.5.21.1
103	Functioning - 1/2 or 5/8 inch Celotex	100	1/2/	Acc-3 Rej-4	3.7	4.5.21.2
104	Non-functioning	10	1/2/	Acc-0 Rej-1	3.8	4.5.22
105	Premature	210	3/	Acc-0 Rej-1	3.9	4.5.23
106	Pitch diameter of thread			0.40%	3.2	Gage
107	Major diameter of thread, max.			0.40%	3.2	Gage
108	True position of large cavity with small cavity			0.40%	3.2	Gage
109	Outside diameter, max.			0.40%	3.2	Gage
110	Total length, max.			0.40%	3.2	Gage
111	Perpendicularity of crimped surface with large outside diameter			0.40%	3.2	Gage

NOTES:
1/ Units subjected to transportation vibration test prior to this test
2/ Air pressure units to be expended in these tests
3/ Test performed concurrently with functioning and non-functioning tests

Source: https://assist.dla.mil-33 Downloaded: 2026-10-26T04:03Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-50864B (AR)

PARAGRAPH	TITLE		SHEET 3 OF 3		DRAWING NUMBER
4.4.2.21	Fuze, PIBD, M549, Less Spitback				9287861
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Minor					
201	Major diameter of thread, min.		0.65%	3.2	Gage
202	Marking, missing, misleading or unidentifiable		0.65%	3.2	Visual
203	Poor workmanship		0.65%	3.12	Visual

NOTES:

AMSMC Form 1570, 1 Feb 85

Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:03Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.22	Unsealed Inner Bag		9287859
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Incorrect number of fuzes		0.40%	3.2	Visual
102	Incorrect number of trays		0.40%	3.2	Visual
103	Fuze inverted		0.40%	3.2	Visual
104	Plastic film missing		0.40%	3.2	Visual
105	Improper strapping		0.40%	3.2	Visual
Minor	None defined				

NOTES:

Source: https://assist.dla.mil - Downloaded: 2026-10-26T04:03Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-50864B (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9287859
4.4.2.23	Sealed Inner Bag				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Heat seal test	1/	1/	3.2	4.5.24
102.	Bag torn or perforated		0.40%	3.2	Visual
103	Bag improperly sealed		0.40%	3.2	Visual
Minor	None defined				

NOTES:
1/ Sampling and rejection shall be in accordance with MIL-P-116.

Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:03Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-50864B (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.24	Sealed, Box, Fiber		9287859
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Contents loose		0.40%	3.2	Visual/Manual
102	Box damaged		0.40%	3.2	Visual
103	Improper sealing		0.40%	3.2	Visual
Minor					
201	Marking misleading or unidentifiable		0.65%	3.2	Visual

NOTES:

AMSMC Form 1570, 1 Feb 85

Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:03Z
Check the source to verify that this is the current version before use.

4.4.3 Testing

4.4.3.1 Check test for deterioration of detonators (see MIL-D-50865). If the total time between original acceptance of any detonator lot and the assembly of that detonator lot into the fuze exceeds two years, or if the detonators have been subjected to adverse conditions, however brief, at any time since previous tests, the detonator lot shall be subjected to and must satisfactorily pass the check test for deterioration specified in MIL-D-50865 immediately before the detonator lot is assembled into the fuze. This test shall be performed by the contractor on detonators selected by the Government inspector at the facility assembling the detonators into the fuze. The test shall be conducted at Government expense without cost to the contractor who loaded the detonator or the contractor assembling the detonator into the fuze and shall not constitute a basis for rejection against either contractor except where deterioration has occurred as a direct result of carelessness in handling, storage, etc., permitted while the detonator lot was under the responsibility of either contractor.

4.4.4 Inspection equipment. The inspection equipment required to perform the examinations and test prescribed herein is described in the Paragraph Reference/Inspection Method column in the tables starting with Paragraph 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and 6.4 herein.

4.5 Test methods and procedures.

4.5.1 Salt spray. This test shall be conducted for 96 hours in accordance with ASTM-B117-73. A 5% salt solution shall be used for the test.

4.5.2 Load requirement at each position. The applicable spring shall be placed in an approved fixture and the load requirement at each position shall be measured. Observation shall be made for compliance with the applicable drawing requirement. This is a non-destructive test. Parts so tested may be returned to the lot.

4.5.3 Push test with simulated setback pin. The setback spring shall be placed in an approved fixture and the force specified on the applicable drawing shall be applied to either a setback pin, Dwg. 9287882, or an equivalent test item. Observation shall be made for compliance with the applicable drawing requirement. This is a destructive test. Setback pins and setback springs subjected to this test shall not be returned to the lot.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:03Z
Check the source to verify that this is the current version before use.

4.5.4 Push-off test of starwheel. The pinion assembly shall be placed in an approved fixture and the force specified on the applicable drawing shall be applied. Observation shall be made for compliance with the applicable drawing requirement. This is a non-destructive test. Assemblies subjected to this test may be returned to the lot.

4.5.5 Torque of starwheel. The pinion assembly shall be placed in an approved fixture and the torque specified on the applicable drawing shall be applied. Observation shall be made for compliance with the applicable drawing requirement. This is a destructive test. Assemblies subjected to this test shall not be returned to the lot.

4.5.6 Push-off of verge arbor. The verge assembly shall be placed in an approved fixture and the force specified on the applicable drawing shall be applied. Observation shall be made for compliance with the applicable drawing requirement. This is a non-destructive test. Assemblies subjected to this test may be returned to the lot.

4.5.7 Push out of firing pin. The top plate assembly shall be placed in an approved fixture and the force specified on the applicable drawing shall be applied. Observation shall be made for compliance with the applicable drawing requirement. This is a non-destructive test. Assemblies subjected to this test may be returned to the lot.

4.5.8 Stake of detonator. The rotor and gear assembly shall be placed in an approved fixture and the force specified on the applicable drawing shall be applied. Observation shall be made for compliance with the applicable drawing requirement. This is a non-destructive test. Assemblies subjected to this test may be returned to the lot.

4.5.9 Push-off test of rotor. The rotor and gear assembly shall be placed in an approved fixture and the force specified on the applicable drawing shall be applied. Observation shall be made for compliance with the applicable drawing requirement. This is a non-destructive test. Assemblies subjected to this test may be returned to the lot.

4.5.10 Push out test of actuator. Prior to the test, the ogive shall be machined to accept the push out feature of the fixture. The modified ogive assembly shall be placed in an approved fixture and the force specified on the applicable drawing shall be applied. Observation shall be made for compliance with the applicable drawing requirement. This is a destructive test. Assemblies subjected to this test shall not be returned to the lot.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:03Z
Check the source to verify that this is the current version before use.

4.5.11 Automatic inspection equipment (redundant). This inspection shall be performed using equipment (See MIL-A-48078) designed by the contractor and approved by the Government. The automatic inspection equipment shall consist of two probes, sensors or transducers, performing the same inspection with automatic transfer between the probes sensors or tranducers, of the assembly being inspected.

4.5.12 Arming time. The escapement assembly, less setback pin and setback spring, shall be placed in an approved fixture and rotated at the speed specified on the applicable drawing. Observation shall be made for compliance with the applicable drawing requirement. This is a non-destructive test. Assemblies subjected to this test may be returned to the lot.

4.5.13 Non-arming test. After successful completion of the test specified in 4.5.12, the escapement assembly, less setback pin and setback spring, shall be reset, placed in an approved fixture and rotated at the non-arming speed specified on the applicable drawing. Observation shall be made for a blocked setback pin hole. This is a non-destructive test. Assemblies subjected to this test may be returned to the lot.

4.5.14 Automatic inspection equipment. This inspection shall be performed using automatic inspection equipment (See MIL-A-48078) designed by the contractor and approved by the Government.

4.5.15 Stake of top plate assembly. The escapement assembly, less the rotor and gear assembly or rotor gear assembly, as applicable, safety spring assembly, drive screw, verge assembly, pinion assembly and anti-creep spring, shall be placed in an approved fixture and the force specified on the applicable drawing shall be applied. Observation shall be made for compliance with the applicable drawing requirement. This is a destructive test. Assemblies subjected to this test shall not be returned to the lot.

4.5.16 X-ray examination of fuze assembly. The fuze assembly shall be x-rayed using approved test equipment and test technique in accordance with MIL-STD-453. Observation shall be made for compliance with the requirements specified in 3.3. This inspection shall be conducted in two (2) planes: from the top of the fuze assembly down; and from the side. The side view shall show the setback pin adjacent to the rotor. After successful completion of this inspection, the fuze assembly shall be subjected to the transportation vibration, jolt or jumble test.

4.5.17 Transportation vibration. The fuze assemblies packed in their regular shipping containers shall be tested in accordance with Test No. 401, Procedure II, of MIL-STD-331. After completion of the test, the fuze assemblies shall be x-rayed as specified in 4.5.16. Observation shall be made for compliance with the requirements specified in 3.4.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:03Z
Check the source to verify that this is the current version before use.

4.5.18 Jolt. The fuze assembly shall be tested in accordance with Test No. 101 of MIL-STD-331. After completion of the test, the fuze assembly shall be x-rayed as specified in 4.5.16. Observation shall be made for compliance with the requirements specified in 3.5.

4.5.19 Jumble. The fuze assembly shall be tested in accordance with Test No. 102 of MIL-STD-331. After completion of the test, the fuze assembly shall be x-rayed as specified in 4.5.16. Observation shall be made for compliance with the requirements specified in 3.6.

4.5.20 Air pressure. After successful completion of the transportation vibration test, the fuze assembly shall be placed in an approved fixture and a measured quantity of air applied to produce the required pressure. Observation shall be made for compliance with the applicable drawing requirement. After successful completion of this test, the fuze assembly shall be subjected to the test specified in 4.5.21.1, 4.5.21.2 or 4.5.22.

4.5.21 Functioning. After successful completion of the transportation vibration test, a spitback assembly (Dwg. 9287862) and spitback cup (Dwg. 12624504) shall be assembled to the fuze, PIBD M549, less spitback (Dwg. 9287861) in accordance with Dwg. 9287860. In turn, this assembly shall be assembled to the test vehicle w/crtg case M169 for M549 fuze (Dwg. 9349828) in accordance with Dwg. 9349971.

4.5.21.1 Coarse sand. The assembled cartridges (see 4.5.21) shall be fired from a M129 launcher or a MK19 Mod 3 machine gun at an elevation that will cause it to impact at approximately 200 meters from the weapon. The impact medium shall be 10 inches, minimum, of course sand which shall be graded at an angle of approximately 25° with the horizontal. The sand shall be raked or the weapon tranversed between shots so that the impacts are not overlapping. If a round falls outside the specified area, it shall be declared a "no-test". When results from the functioning test are reported, the number of "no-tests" shall also be reported. Observation shall be made for compliance with the requirements specified in 3.7.

4.5.21.2 One-half (1/2) or five-eights (5/8) inch celotex. The assembled cartridge (see 4.5.20) shall be fired from a M129 launcher or MK19 Mod 3 machine gun to impact at approximately 200 feet against a vertical 1/2 or 5/8 inch piece of Celotex. If a round fails to hit the Celotex, it shall be declared a "no-test". When the results from the functioning test are reported, the number of "no-tests" shall also be reported. Observation shall be made for compliance with the requirements specified in 3.7.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:03Z
Check the source to verify that this is the current version before use.

4.5.22 Non-functioning. After successful completion of the transportation vibration test, a spitback assembly (Dwg. 9287862) and spitback cup (Dwg. 12624504) shall be assembled to the fuze, PIBD M549, less spitback (Dwg. 9287861) in accordance with Dwg. 9287860. In turn, this assembly shall be assembled to the test vehicle w/crtg case M169 for M549 fuze (Dwg. 9349828) in accordance with Dwg. 9349971. The assembled cartridge shall be fired from a M129 launcher or MK19 Mod 3 machine gun to impact at 60 feet against a vertical 1/2 or 5/8 inch thick piece of Celotex. If a round fails to hit the target or hits the target and does not function and does not function on impact on a vertical 1/2" thick mild steel plate placed at 200 feet from the launcher, it shall be declared a "no-test". When results from the non-functioning test are reported, the number of "no-tests" shall also be reported. Observation shall be made for compliance with the requirement specified in 3.8.

4.5.23 Premature burst. This test shall be conducted concurrently with the tests specified in 4.5.21 and 4.5.22. Observation shall be made for compliance with the requirement specified in 3.9.

4.5.24 Heat seal test of bag. The test shall be performed in accordance with the procedure specified in MIL-P-116.

5 PACKAGING.

5.1 Preservation and packaging.

5.1.1 Level B. The fuze shall be preserved and packaged in accordance with Dwg. 9287859.

5.2 Packing.

5.2.1 Level B. The fuze, preserved and packaged as specified in 5.1.1, shall be packed for shipment in accordance with Dwg. 9287859.

5.3 Marking. Marking shall be in accordance with Dwg. 9287859 and Code of Federal Regulations, Title 49, Parts 100-199.

5.4 Shipping. When fuze assemblies from more than one lot are shipped as a carload, each lot shall be kept separate and the division between the lots clearly indicated to prevent mixing of the lots in transit.

6. NOTES

6.1 Intended use. The fuze assemblies covered by this specification are intended for use with the Cartridge, 40mm, HEDP, M430.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:03Z
Check the source to verify that this is the current version before use.

6.2 Ordering data. Ordering data should include:

a. Title, number and date of specification.

b. Quantity required, delivery schedules and cartridge and weapon system that fuzes are being procured for.

c. Quality Conformance Inspection, if other than specified in Section 4 of this specification.

d. First Article Sample requirements, if other than specified in Section 4 of this specification.

e. Packaging requirements, if other than that specified in Section 5 of this specification.

f. Contractor's quality program. Contracts should specify MIL-Q-9858 as the applicable higher level contract quality control requirement.

6.3 Data requirements. When this specification is used in an acquisition and data are required to be delivered, the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (CDRL), incorporated into the contract. When the provisions of DOD FAR supplement, Part 27, Sub-part 27.410-6 (DD Form 1423) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data requied by this specification are cited in the following paragraphs.

Reference Para	DID No	DID Title	Suggested Tailoring
3.11	DI-R-5297	Quality Program Plan	Apply material control and process control provisions

(Data item descriptions related to this specification and identified in section 6 will be approved and listed as such in DOD 5000.19L, Vol II, AMSDL. Copies of data item descriptions required by the contractors in connection with specified acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.4 Submission of inspection equipment designs for approval. Submit equipment designs as required to Commander: US Army Armament Munitions and Chemical Command, ATTN: AMSMC-QAF-I (D), Picatinny Arsenal, New Jersey 07806-5000. Request letter of submittal shall state contractor, contract number, specification number, item nomenclature and classification of defects or test.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:03Z
Check the source to verify that this is the current version before use.

6.5 Changes from previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

6.6 Drawings. Drawings listed in Section 2 of this specification under the headings US Army Research, Development and Engineering Center (ARDEC) may also include drawings prepared by and indentified as, Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data orignially prepared by these activities is now under the cognizance of ARDEC.

6.7 Subject term (key words) listing.

Cartridge, 40mm, HEDP, M430
Cup, Spitback
Detonator, Stab, M59
Gun, Machine, 40mm, MK19 Mod 3
Launcher, Grenade, 40mm: M129

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 1390-A562)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:03Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE ARMY

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 12062 WASHINGTON D. C.
POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE ARMY

COMMANDER
US ARMY ARDEC
ATTN: AMSMC-QA
PICATINNY ARSENAL, NJ 07806-5000

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:03Z
Check the source to verify that this is the current version before use.

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-F-50864B	Fuze, PIBD, M549; LESS SPITBACK METAL PARTS & LAP

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:03Z
Check the source to verify that this is the current version before use.